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Development and Standardization of an Osteoradionecrosis Classification System in Head and Neck Cancer: Implementation of a Risk-Based Model

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DOI https://doi.org/10.1200/JCO.23.01951

ABSTRACT

PURPOSE Osteoradionecrosis of the jaw (ORN) can manifest in varying severity. The aim of this study is to identify ORN risk factors and develop a novel classification to depict the severity of ORN.

METHODS Consecutive patients with head and neck cancer (HNC) treated with curative-intent intensity-modulated radiation therapy (IMRT) (≥45 Gy) from 2011 to 2017 were included. Occurrence of ORN was identified from in-house prospective dental and clinical databases and charts. Multivariable logistic regression model was used to identify risk factors and stratify patients into high-risk and low-risk groups. A novel ORN classification system was developed to depict ORN severity by modifying existing systems and incorporating expert opinion. The performance of the novel system was compared with 15 existing systems for their ability to identify and predict serious ORN event (jaw fracture or requiring jaw resection).

RESULTS ORN was identified in 219 of 2,732 (8%) consecutive patients with HNC. Factors associated with high risk of ORN were oral cavity or oropharyngeal primaries, received IMRT dose ≥60 Gy, current/ex-smokers, and/or stage III to IV periodontal condition. The ORN rate for high-risk versus low-risk patients was 12.7% versus 3.1% ($P < .001$) with an AUC of 0.71. Existing ORN systems overclassified serious ORN events and failed to recognize maxillary ORN. A novel ORN classification system, ClinRad, was proposed on the basis of vertical extent of bone necrosis and presence/absence of exposed bone/fistula. This system detected serious ORN events in 5.7% of patients and statistically outperformed existing systems.

CONCLUSION We identified risk factors for ORN and proposed a novel ORN classification system on the basis of vertical extent of bone necrosis and presence/absence of exposed bone/fistula. It outperformed existing systems in depicting the seriousness of ORN and may facilitate clinical care and clinical trials.

ACCOMPANYING CONTENT

Appendix

Data Supplement

Accepted February 26, 2024
Published May 1, 2024

J Clin Oncol 42:1922-1933
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INTRODUCTION

Osteoradionecrosis (ORN) is one of the most severe radiation-induced complications for patients with head and neck cancer (HNC). ORN has been loosely defined as non-healing mucosal breakdown and bone injury, occurring spontaneously or after trauma, within the head and neck (HN) radiation volume in the absence of recurrent tumor on the affected site. Although these criteria are widely accepted for a clinical diagnosis, they fail to incorporate objective imaging findings.[1]

The prevalence of ORN varies widely in the literature (up to 40%) with a substantial decrease in modern era (4%-8%),[2] attributed to advances in radiotherapy technology[3,4] and the use of a strict prophylactic dental care policy among institutions.[5] Specific ORN events, in particular ORN-related jaw fracture, are associated with high resource utilization at health care system level and significant negative impact on the quality of life of patients with HNC, influencing patients' subjective well-being on physical, emotional, functional, and social levels.[6] In addition to other indications, ORN-related pathologic fracture

CONTEXT

Key Objective

To develop a novel osteoradionecrosis (ORN) staging and grading system that is applicable clinically and in a trial setting.

Knowledge Generated

The ClinRad classification system for ORN considers clinical and radiographic features to stage the disease in clinical practice and considers treatment recommendations to grade the disease in trials setting. A high-risk cohort for ORN was identified and consists of patients with oral cavity or oropharyngeal primaries who received IMRT dose ≥60 Gy and who are current/ex-smokers and have a stage III to IV periodontal condition.

Relevance *(J.P.S. Knisely)*

ORN consequent to chemoradiotherapy for head and neck cancer is justifiably feared. Along with confirmation of the prognostication scheme on a separate clinical database, steps to lessen the risk of developing ORN should also be initiated and tested to make this complication vanishingly rare.*

*Relevance section written by *JCO* Associate Editor Jonathan P.S. Knisely, MD.

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necessitates surgical intervention, a significant additional health care system burden resulting in lengthier hospitalization periods, treatment costs, and financial burden for patients.[7] Regardless of the indication, major surgical interventions represent a serious event from both patient and health care system perspectives.[8]

The severity of ORN is depicted by various classification systems, generally defined on the basis of (1) clinical signs (eg, exposed bone), (2) radiographic findings (eg, pathologic fracture), and (3) treatment required and response to therapy (either conservative or surgical). However, to the best of our knowledge, there currently is no consensus regarding the most appropriate classification system, which hinders interstudy comparison and clinical trial design. Furthermore, the most well-accepted systems overclassify serious ORN,[9-12] fail to acknowledge maxillary ORN,[13-16] or rely on subjective findings.[17-20]

Therefore, we conducted this single-institution study aimed to identify risk factors for ORN in patients with HNC treated with intensity-modulated radiation therapy (IMRT), evaluate the appropriateness of various ORN classification systems in depicting the severity of ORN, and propose a reproducible and standardized novel ORN classification system incorporating clinically relevant data that can be used in routine clinical care and future clinical trials.

METHODS

Study Population and Data Compilation

We conducted a retrospective review of all consecutive adult patients with HNC treated with curative-intent (definitive, preoperative, or postoperative) IMRT ≥ 45 Gy in our institution who underwent preradiation dental evaluation at our institutional dental oncology clinic from 2011 to 2017. Patients who did not undergo preradiotherapy (RT) dental assessment at our institution, those with a previous history of HN radiation, a diagnosis of primary tracheal cancer or early-stage laryngeal cancer (T1-2 N0 M0), or edentulous patients were excluded. There was no exclusion on the basis of adequacy of available data. A flow diagram is included as Appendix Figure A1 (online only). Institutional research ethics approval was obtained before study initiation.

Clinical information, and incidence and severity of ORN were retrieved from in-house prospective *Clinical* and *Dental* databases, and cross-checked against individual patient dental charts to ensure the completeness and quality of the data. A full description of the databases is included in Appendix Figure A1.[21,22,23]

Assessment and Classification of ORN Severity

The criteria used to define an ORN event were adapted from previous classification systems and included clinical signs, radiographic findings, treatment required, and response to therapy. Indications for jaw resection in patients with ORN at our institution included jaw fracture, persistent draining fistula, and significant exposed bone with basilar bone involvement after failure of conservative management in symptom control. Each individual visit for all patients with ORN was reviewed and ORN events were retrospectively classified according to 15 clinically used ORN classification systems reported in the literature (Appendix Table A1): Coffin,[9] Marx,[20] Morton and Simpson,[10] Radiation Therapy Oncology Group/European Organisation for Research and Treatment of Cancer (RTOG/EORTC),[24] Glanzmann and Grätz,[17] Clayman,[11] Late Effects Normal Tissue Task Force Subjective, Objective, Management, Analytic scales

(LENT/SOMA),[25] Støre and Boysen,[26] Schwartz and Kagan,[13] Notani et al,[14] Tsai et al,[18] Karagozoglu et al,[15] Lyons et al,[19] Common Terminology Criteria for Adverse Events (CTCAE),[27] Caparrotti et al,[12] and Shaw et al.[16] For systems that have a duration component (such as the Caparrotti system), duration was used to appropriately grade the ORN according to each classification system. All data related to ORN were independently reviewed by two reviewers.

Distributions of classifications of ORN for each system were summarized across all visits. Proportions of visits *missing* sufficient data to assign a classification in each system were summarized and presented both unweighted (all visits weighted equally) and weighted (mean proportion of missing visits was first calculated for each patient, then aggregated so that each patient contributed equal weight). Proportions of patients with at least one classification in the *most severe category* for each system were summarized.

Statistical Analysis

Identifying ORN Risk Factors and Stratifying Risk Groups

Univariable analysis with logistic regression was conducted to assess the association of the following clinical factors with ORN, chosen on the basis of clinical relevance: primary tumor location, stage, sex, smoking status, alcohol intake, human papillomavirus status, IMRT dose, concurrent chemotherapy, surgical management, periodontal condition, age at diagnosis, and smoking pack/years. Variables with $P < .1$ in univariable analysis were considered for inclusion in a multivariable ORN risk score model, built using backward model selection, which classified patients into high- and low-risk groups. Total risk scores were computed by adding the resulting model coefficients on the log scale (termed risk score coefficients [RSC]). An optimal cutpoint for classifying patients as high-risk versus low-risk was identified by maximizing Youden's J.[28] To measure predictive performance of the binary high-risk versus low-risk scores, AUC was calculated from repeated 10-fold cross-validation with 10 iterations. All tests were two-sided and used $P < .05$ to define statistical significance.

Evaluating the Performance of ORN Classification Systems

The performance of ORN classification systems were evaluated for *separability*, defined here as the ability of each system to discriminate between patients who would go on to experience a serious ORN event (jaw fracture or ORN requiring jaw resection) versus those who would not. First, Kaplan-Meier curves and log-rank tests were used to compare time from ORN detection to serious ORN event, stratified by stage/grade at the time of ORN detection. Second, univariable Cox proportional hazards regression models were fit to evaluate time to serious ORN event from ORN stage/grade at any visit after detection of ORN, treating ORN stage as a time-varying covariate. Finally, predictive performance of each classification system was evaluated via overall Cox model concordance, as well as time-dependent AUC at several clinically relevant time points and corresponding c-indices using the methods described by Bansal and Heagerty.[29] Time-dependent AUC measures classification accuracy between patients with a serious ORN event versus no events at a given landmark time. Separability analysis was not performed on systems for which >50% of all visits were missing/nonevaluable. A full description of AUC analysis is found in the Data Supplement (online only).[29,30] We considered concordance and c-index statistics above 0.7, 0.8, and 0.9 to indicate relevant, excellent, and outstanding predictive performance, respectively.

Proposing a Novel ORN Classification System

The existing ORN classification systems used to inform the novel classification system were those identified as high-performing on our analysis on the basis of proportion of missingness (and most severe stage or grade) and separability (time to serious negative ORN outcomes assessed by Kaplan-Meier curves from first ORN detection, time-to-event Cox regression treating ORN stage as a time-varying covariate, and time-dependent AUC). Final extracted characteristics included (1) the Notani et al[14] system: categorization of early stage ORN as confined to alveolar bone on the basis of imaging; (2) Shaw et al[16] system: addition of minor bony spicules; (3) Støre and Boysen[26] system: inclusion of radiographic ORN with intact mucosa; (4) CTCAE[27]: aligning objective grading criteria with treatment recommendations, and; (5) multiple systems that recognize fracture and oro-cutaneous fistula as representing the most severe form of ORN.[10,12-14,16-20,24-26] In addition, The American Association of Oral and Maxillofacial surgeons (AAOMS) staging system for medication-related osteonecrosis of the jaw (MRONJ)[31,32] was used, in particular the recognition of ORN detectable radiographically but not clinically.

RESULTS

Clinical Characteristics

A total of 2,732 patients with HNC were included. Median age was 61 years; 30% (n = 807) were current/ex-smokers; 53% (n = 1,410) had stage III to IV periodontal condition (PC); 18% (n = 490) were oral cavity cancer (OCC), and 35% (n = 969) had oropharyngeal cancer (OPC). The median IMRT dose was 70 Gy/35 fractions; 43% (n = 1,170) received concurrent chemotherapy and 37% (n = 1,006) underwent surgery (Table 1). The median follow-up was 5.6 years (IQR, 4.4-7.3 years). A total of 219 patients (8%) developed ORN.

ORN Risk Factors and Risk Groups

Multivariable analysis (Table 2) identified the following ORN risk factors: current/ex-smoker (RSC = 0.4), stage III to IV PC (RSC = 0.4), primary OCC (RSC = 1.7), primary OPC (RSC = 1.4), and IMRT dose prescription ≥60 Gy (RSC = 1.3). Patients were classified into ORN risk groups according to

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TABLE 1. Patient, Tumor and Treatment Characteristics

Covariate	Full Sample (N = 2,732)	ORN (no) (n = 2,513)	ORN (yes) (n = 219)	*P*
Insurance type, No. (%)				**<.001**
No insurance	1,160 (42)	1,080 (43)	80 (38)	
Private	1,373 (50)	1,264 (50)	109 (49)	
Public	199 (7)	169 (7)	30 (14)	
Teeth preRT, median (range)	25 (1-32)	25 (1-32)	25 (2-32)	.60
Teeth postRT, median (range)	24 (0-32)	24 (0-32)	23 (0-32)	.35
DMFS, median (range)	71 (0-160)	71 (0-160)	72 (5,160)	.86
Smoking, No. (%)				**<.001**
No	1,925 (70)	1,794 (71)	131 (60)	
Yes	807 (30)	719 (29)	88 (40)	
Drinking, No. (%)				**.006**
Never-drinker	1,038 (38)	973 (39)	65 (30)	
Ever-drinker	1,528 (56)	1,383 (55)	145 (66)	
Unknown	166 (6)	157 (6)	9 (4)	
Chemotherapy, No. (%)				.16
No	1,563 (57)	1,448 (58)	115 (54)	
Yes	1,169 (43)	1,065 (42)	104 (46)	
Surgery, No. (%)				.76
No	1,727 (63)	1,586 (63)	141 (63)	
Yes	1,005 (37)	927 (37)	78 (37)	
Age at Dx, years, median (range)	61.0 (18.3-96.7)	61.3 (18.3-96.7)	59.0 (27.7-88.9)	.082
Sex, No. (%)				.77
Female	732 (27)	671 (27)	61 (28)	
Male	2,000 (73)	1,842 (73)	158 (72)	
N category, No. (%)				.08
N0	790 (29)	726 (29)	64 (29)	
N1	351 (13)	325 (13)	26 (12)	
N2	1,405 (52)	1,282 (52)	123 (56)	
N3	156 (6)	151 (6)	5 (2)	
Missing	30	29	1	
Disease site, No. (%)				**<.001**
Other	1,074 (39)	1,039 (41)	35 (16)	
Larynx	199 (7)	197 (8)	2 (1)	
Lip and oral cavity	490 (18)	417 (17)	73 (33)	
Oropharynx	969 (35)	860 (34)	109 (50)	
Human papillomavirus status, No. (%)				**.06**
Negative	448 (16)	408 (16)	40 (18)	
Positive	856 (31)	775 (31)	81 (37)	
Unknown/not tested	1,428 (52)	1,330 (53)	98 (45)	
Teeth removed, No. (%)				.29
Mean (SD)	1.3 (3.0)	1.3 (2.9)	1.6 (3.6)	
Median (min, max)	0 (0, 28)	0 (0, 26)	0 (0, 28)	
Missing	1	1	0	
Proportion teeth removed, No. (%)				.33
Mean (SD)	0.1 (0.2)	0.1 (0.2)	0.1 (0.2)	
Median (min, max)	0 (0, 1)	0 (0, 1)	0 (0, 1)	
Missing	1	1	0	

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TABLE 1. Patient, Tumor and Treatment Characteristics (continued)

Covariate	Full Sample (N = 2,732)	ORN (no) (n = 2,513)	ORN (yes) (n = 219)	*P*
Smoking PY, median (range)	26 (1-89)	21 (1-89)	35 (1-86)	**<.001**
PC, No. (%)				**.012**
I	270 (10)	252 (10)	18 (8)	
II	504 (19)	467 (19)	37 (17)	
III	747 (28)	673 (27)	74 (33)	
IV	663 (25)	599 (24)	64 (30)	
O	489 (18)	464 (19)	25 (12)	
Missing	59	58	1	
Pre-RT extraction, No. (%)				.38
No	1,776 (65)	1,639 (65)	136 (62)	
Yes	956 (35)	873 (35)	83 (38)	
RT dose, median (range)	70.0 (45-76.7)	70.0 (45-76.7)	70 (50-70)	.16

NOTE. Bold values denote a significant result.
Abbreviations: DMFS, decayed, missing, filled surfaces according to DMFS160; ORN, osteoradionecrosis; PC, periodontal condition; Post-RT, after completion of radiotherapy; Pre-RT, before commencement of radiotherapy; PY, pack-years.

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the sum of their RSC (high-risk group: sum of RSC ≥3.5; low-risk group: sum of RSC <3.5). Applying the generated model, the high-risk group included patients with OCC or patients with OPC who received IMRT ≥60 Gy and current/ex-smoker with stage III to IV PC. The overall rate of ORN for high-risk patients was 12.7% (95% CI, 11 to 14.5) versus 3.1% (95% CI, 2.1 to 4) in the low-risk group ($P < .001$). Model performance from 10-fold cross-validation showed an AUC of 0.71 (95% CI, 0.61 to 0.81).

Comparison of ORN Classification Systems

Missingness (the average proportion of visits where an ORN stage could not be generated per patient by each grading system) is summarized in Table 3. The following ORN grading systems showed <25% missingness: Clayman[11] (5.4%), CTCAE[27] (20.6%), Notani et al[14] (20.9%), RTOG/EORTC[24] (19.3%), and Støre and Boysen[26] (7.5%) systems. The proportion of patients with at least one most severe stage is summarized in Table 3. The following ORN classification systems detected <5% of patients as having the most severe stage or grade of ORN: RTOG/EORTC[24] (1.1%), SOMA[25] (4.7%), Støre and Boysen[26] (4.9%), and Schwartz and Kagan[13] (2.5%) systems.

TABLE 2. Multivariable Analysis of Risk Factors for Osteoradionecrosis

Risk Factor	Odds Ratio (95% CI)	*P*	Global *P*	N
Smoking		**.010**		2,673
No	Reference			1,896
Yes	1.48 (1.10 to 1.99)			777
Primary site			**<.001**	2,673
Larynx/other	Reference			1,246
Oral cavity	5.66 (3.74 to 8.55)	**<.001**		476
Oropharynx	4.05 (2.75 to 5.96)	**<.001**		951
Dose		**.03**		2,673
<60	Reference			144
≥60	3.59 (1.12 to 11.51)			2,529
Periodontal classification		**.004**		2,673
0-2 (mild)	Reference			1,263
3-4 (moderate-severe)	1.55 (1.15 to 2.08)			1,410

NOTE. Bold values denote a significant result.

Separability of different classification systems is illustrated in Figure 1. A total of 168 patients had sufficient ORN staging and follow-up data for separability analysis, and 32 patients had a serious ORN event (ie, jaw fracture or surgery). For the Notani et al,[14] Shaw et al,[16] (stages II to III grouped), RTOG/EORTC (grades 2 to 4 grouped), Støre and Boysen[26] (grades 2 to 3 grouped), Schwartz et al,[13] and Karagozoglu et al[15] (grades 1 to 3 grouped) systems, patients with a more severe grade at the time of ORN detection were more likely to develop a serious ORN event (fracture or surgery) at an earlier time point than patients with each successively less severe grade. On univariable Cox regression (Table 4), diagnosis with most severe stages of the Notani (stages II to III), Shaw (stages II to III), and RTOG/EORTC (grades 2 to 4) systems significantly associated with risk of serious ORN event (fracture or surgery) when compared with patients only diagnosed with the least severe stage/grade(s).

Development and Evaluation of Novel ORN Grading System

The proposed ORN classification system categorizes ORN on the basis of vertical extent of necrosis on imaging and objective clinical findings of bone exposure or fistula formation and is named ClinRad to reflect reliance on objective clinical and imaging findings. The system recognizes four different Stages of ORN in addition to minor bony spicules (Table 5):

TABLE 3. Count and Proportion of Visits Missing Staging Information in Each System and Proportion of All Patients With at Least One Visit With Most Severe Stage

Classification System	Total Visits	No. of Staged Visits	No. of Nonevaluable[a] Visits	Proportion of All Visits Nonevaluable	Average Proportion Missing Per Patient[b]	Most Severe Stage	Proportion of Patients With at Least One Most Severe Score	Proportion All Patients With at Least One Most Severe Score, Excluding Patients With Only Nonevaluable Stages
Caparrotti et al[12]	1,063	626	437	0.411	0.423	4	0.151	0.182
Clayman[11]	1,063	989	74	0.070	0.054	2	0.835	0.847
Coffin[9]	1,063	202	861	0.810	0.783	Major	0.057	0.152
CTCAE[27]	1,063	867	196	0.184	0.206	3	0.057	0.066
Glanzmann and Grätz[17]	1,063	379	684	0.643	0.794	2	0.165	0.449
Karagozoglu et al[15]	1,063	637	426	0.401	0.434	3	0.061	0.088
Lyons et al[19]	1,063	355	708	0.666	0.712	4	0.071	0.176
Marx[20]	1,063	160	903	0.849	0.926	3	0.024	0.185
Morton and Simpson[10]	1,063	170	893	0.840	0.800	Major	0.057	0.164
Notani et al[14]	1,063	878	185	0.174	0.209	3	0.113	0.132
ClinRad	**1,063**	**1,063**	**0**	**0.000**	**0.000**	**3**	**0.057**	**0.057**
RTOG/EORTC[24]	1,063	872	191	0.180	0.193	4	0.009	0.011
Schwartz and Kagan[13]	1,063	728	335	0.315	0.323	3b	0.019	0.025
SOMA[25]	1,063	763	300	0.282	0.313	3	0.075	0.116
Støre and Boysen[26]	1,063	983	80	0.075	0.065	4	0.038	0.047
Shaw et al[16]	1,063	574	489	0.460	0.501	3	0.047	0.049
Tsai et al[18]	1,063	699	364	0.342	0.362	4	0.042	0.057

NOTE. Bold indicates a new system.

Abbreviations: CTCAE, Common Terminology Criteria for Adverse Events; EORTC, European Organisation for Research and Treatment of Cancer; RTOG, Radiation Therapy Oncology Group; SOMA, Subjective, Objective, Management, Analytic scales.

[a]Nonevaluable: missing sufficient data to determine stage from medical chart review.

[b]Average proportion of visits where no stage could be generated, per patient.

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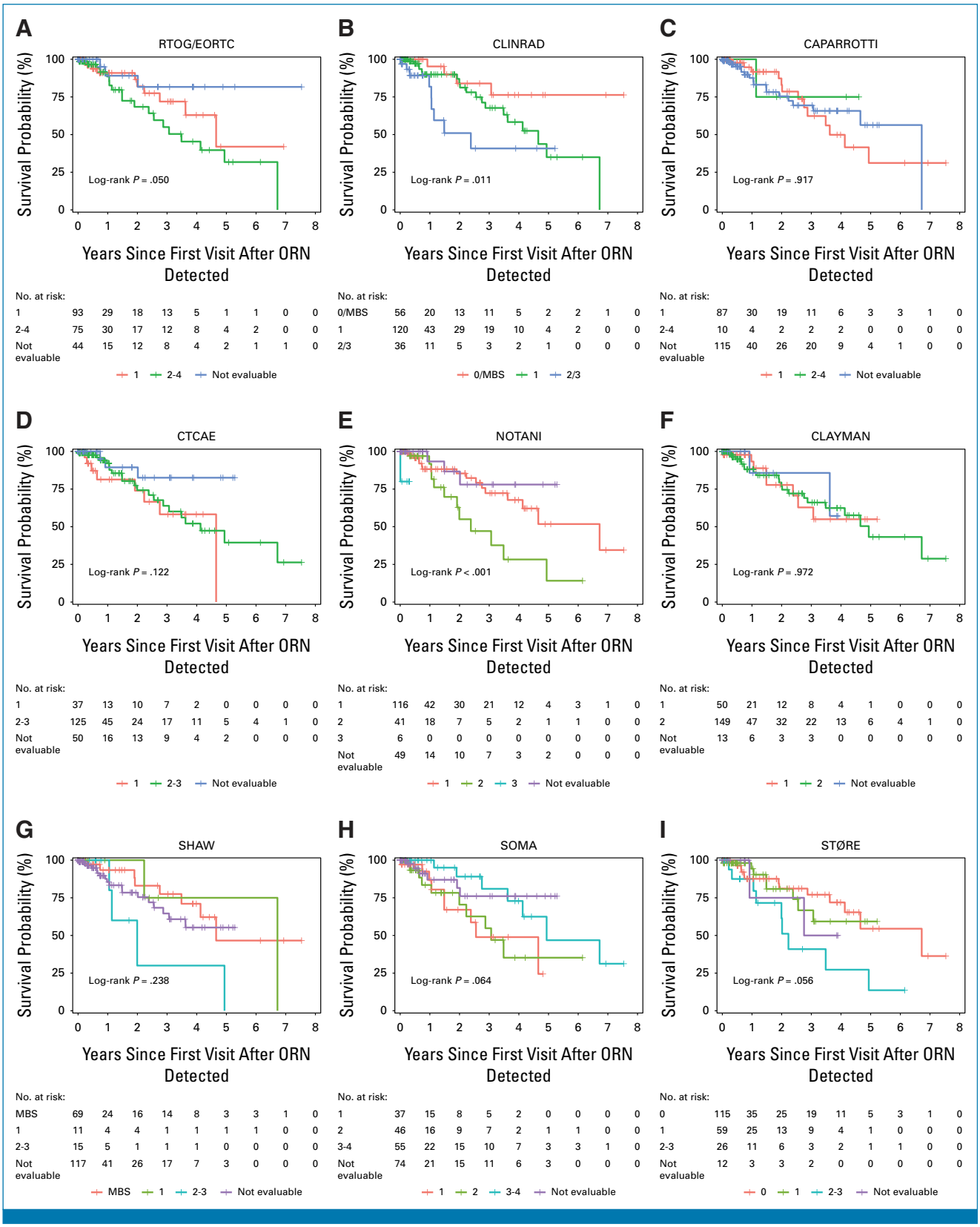

FIG 1. Kaplan-Meier curves for time to jaw fracture or jaw surgery, stratified by stage or grade at baseline (time of ORN detection) within each classification system: (A) RTOG/EORTC,[24] (B) ClinRad, (C) Caparrotti et al,[12] (D) CTCAE,[27] (E) Notani et al,[14] (F) Clayman,[11] (G) Shaw et al,[16] (H) SOMA,[25] and (I) Støre and Boysen.[26] EORTC, European Organisation for Research and Treatment of Cancer; MBS, minor bony spicules; ORN, osteoradionecrosis; RTOG, Radiation Therapy Oncology Group; SOMA, Subjective, Objective, Management, Analytic scales.

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TABLE 4. Univariable Cox Time-To-Event Regression Models: Time to Jaw Fracture or Surgery by Stage at all Visits Before Fracture, as a Time-Varying Covariate

Covariate	HR (95% CI)	*P*	Concordance (95% CI)
ClinRad			0.73 (0.64 to 0.83)
MBS/0/1	Reference		
2	4.90 (2.12 to 11.31)	<.001	
3	11.46 (3.34 to 39.32)	<.001	
Notani et al[14]			0.69 (0.58 to 0.81)
1	Reference		
2	2.50 (1.02 to 6.14)	.05	
3	9.85 (3.51 to 27.69)	<.001	
Shaw et al[16]			0.67 (0.55 to 0.79)
MBS-1	Reference		
2	2.54 (0.97 to 6.64)	.06	
3	7.02 (2.47 to 19.96)	<.001	
Caparrotti et al[12]			0.67 (0.56 to 0.78)
1	Reference		
2	1.07 (0.19 to 5.94)	.94	
3	2.75 (0.85 to 8.92)	.09	
4	5.79 (1.67 to 20.04)	.006	
CTCAE[27]			0.64 (0.54 to 0.73)
1	Reference		
2	1.29 (0.54 to 3.08)	.57	
3	13.88 (4.19 to 45.92)	<.001	
RTOG/EORTC[24]			0.70 (0.62 to 0.78)
0-1	Reference		
2	2.67 (1.16 to 6.14)	.02	
3-4	12.99 (3.78 to 44.65)	<.001	
SOMA[25]			0.64 (0.51 to 0.76)
0-1	Reference		
2	1.85 (0.64 to 5.37)	.26	
3	1.02 (0.35 to 3.01)	.97	
4	25.60 (6.37 to 102.91)	<.001	
Støre and Boysen[26]			0.67 (0.55 to 0.79)
0	Reference		
1	1.46 (0.52 to 4.09)	.47	
2	2.05 (0.78 to 5.43)	.15	
3	24.85 (7.22 to 85.52)	<.001	
Schwartz and Kagan[13]			0.67 (0.55 to 0.79)
0-1	Reference		
2	1.87 (0.78 to 4.48)	.16	
3	7.31 (2.50 to 21.36)	<.001	
Tsai et al[18]			0.65 (0.53 to 0.77)
1	Reference		
2	0.57 (0.22 to 1.46)	.24	
3	1.40 (0.44 to 4.44)	.57	
4	15.31 (4.80 to 48.82)	<.001	
Karagozoglu et al[15]			0.60 (0.46 to 0.73)
0	Reference		
1	1.19 (0.30 to 4.66)	.80	
2	1.65 (0.45 to 6.09)	.45	
3	4.43 (1.01 to 19.48)	.05	

Abbreviations: CTCAE, Common Terminology Criteria for Adverse Events; EORTC, European Organisation for Research and Treatment of Cancer; HR, Hazard Ratio; MBS, minor bony spicules; RTOG, Radiation Therapy Oncology Group; SOMA, Subjective, Objective, Management, Analytic scales.

Applying the ClinRad system, we were able to classify all visits with ORN (0.0% missingness) and detected 5.7% of patients with the most severe stage of ORN. Patients with a more severe stage at the time of ORN detection were more likely to develop serious ORN event (fracture or requiring surgical resection) at an earlier time point than patients with each successively less severe stage (Fig 1), with the caveat that stages II and III were grouped for statistical purpose because of low numbers of patients with stage III ORN at the time of ORN detection. In addition, diagnosis of stage II ORN (hazard ratio [HR], 4.9 [95% CI, 2.12 to 11.31]; $P < .001$) or stage III ORN (HR, 11.45 [95% CI, 3.34 to 39.32]; $P < .001$) significantly associated with risk of serious ORN event when compared with patients only diagnosed with the least severe stages. The ClinRad system demonstrated good concordance in Cox modeling, with the highest concordance statistic across all systems (0.73, 95% CI, 0.64 to 0.83), although confidence intervals overlapped with concordance results from other systems (Table 4).

Results of time-dependent AUC analysis are presented in the Data Supplement. Although no system met the threshold for good performance with respect to c-indices, the ClinRad system had numerically the highest c-index of all classification systems in time-dependent AUC analysis, though there was overlap in confidence intervals of the ClinRad system versus other commonly used systems such as Notani and Shaw. The ClinRad system performed similarly or better than other grading systems when predicting serious ORN events at 1, 2, and 3 years after detection of ORN.

DISCUSSION

This single-institution study identified a high-risk group for ORN. A novel ORN classification system, ClinRad, is proposed to address the limitations of existing systems. The ClinRad system also harmonizes with the CTCAE classification to produce a grading system that can be used in clinical trials.

The overall rate of ORN at the current study was 8%, slightly higher than that reported in the IMRT era. This can be attributed to several factors, including (1) inclusion of minor

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TABLE 5. ClinRad—A Novel Classification System for ORN

Description	Stage	Radiographic Findings	Clinical Findings	Intervention	ClinRad Clinical Trial Grade Proposal
Distinct from ORN, occurs more often in patients with history of RT	Minor bone spicules	None, aside from superficial sequestra	Superficial mobile spicules/ sequestra within mucosa	None indicated	Grade 0
Radiographic evidence of bone necrosis confined to alveolar bone with no clinical signs of ORN	Stage 0	Bone necrosis confined to alveolar bone including: Bone lysis/sclerosis Widening periodontal ligament (PDL) space Absence of osseous filling of extraction sockets	Intact mucosa	None indicated	Grade 1
Clinical signs of ORN with or without radiographic evidence of bone necrosis confined to alveolar bone	Stage I	None or as stage 0	Exposed bone[a]	Minor surgical intervention[b] and/or medical management[c] may be indicated with or without adjunctive conservative management[d]	Grade 2
Radiographic evidence involving basilar bone with or without clinical signs of ORN	Stage II	Bone necrosis involving the basilar bone or maxillary sinus	Intact mucosa or exposed bone[a]	Intermediate surgical intervention may be indicated[e], with or without adjunctive conservative and medical management	Grade 3
Advanced ORN	Stage III	One or more of the following: Pathologic fracture Orocutaneous fistula Oral antral communication/oral nasal communication	One or more of the following: Pathologic fracture Orocutaneous fistula Oral antral communication/ oral nasal communication	Reconstructive surgical intervention is indicated,[f] with or without adjunctive conservative and medical management	Grade 4

Abbreviations: ORN, osteoradionecrosis; RT, radiotherapy.
[a]Including pinpoint mucosal breach (intraoral fistula) that probes to bone and/or probing to bone along periodontal tissues.
[b]Examples include sequestrectomy.
[c]Examples include pentoxifylline-tocopherol and hyperbaric oxygen.
[d]Examples include chlorhexidine rinses and antibiotics.
[e]Examples include transoral surgical intervention and debridement, alveolectomy, and soft tissue closure.
[f]Examples include segmental maxillectomy/mandibulectomy with vascularized free tissue reconstruction.

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bony spicules and early radiographic changes as ORN events in this analysis; (2) the utilization of prospectively maintained databases, contrary to the majority of ORN studies that are retrospective in nature and therefore inherently prone to underestimating rates of disease; (3) the detection of ORN events on the basis of multidisciplinary assessments including radiation, surgical, and dental oncologists; and (4) inclusion of medical and dental imaging to capture ORN events that might otherwise go undetected.

With a renewed interest in hypofractionation for HNC, risk models for ORN on the basis of conventional radiotherapy dose parameters cannot be directly applied to hypofractionated or stereotactic body radiation therapy (SBRT). Verduijn et al 2023 performed a retrospective study in 334 patients with HNC treated with either SBRT boost or conventional IMRT.[33] SBRT and IMRT dose distributions were converted into equivalent dose in 2 Gy fractions (a/b = 0.85 Gy) and analyzed using equivalent uniform doses. In multivariable analysis, equivalent uniform dose (a = 8) was significantly associated with ORN (AUC = 0.71, $P < .001$). Therefore, equivalent uniform dose can be incorporated into ORN risk prediction models across different fractionation schemes.[33]

The current analysis excluded edentulous patients because at our institution, edentulous patients are not routinely referred to dental oncologists before RT. However, this does not mean the ClinRad system can only be applied to dentate regions or patients. Some patients seen for preradiation dental assessment were rendered edentulous by the time of radiation. In addition, partially dentate patients with edentulous spans and ORN developing in an edentulous regions were included. In these cases, diagnosis of ORN depends on the stage of resorption of the tooth socket in the edentulous patient or edentulous span, where it is still possible to assign bone exposure or radiographic bone lysis as occurring within alveolar bone or basilar bone.

When considering the performance of previously proposed systems, the Notani system performed well across multiple domains. A major limitation of the Notani system is the inclusion of bone lysis beyond the inferior alveolar canal as stage III disease. This means a patient with longstanding radiographic evidence of bone lysis beyond this landmark who then progresses to fracture will not see an advancement in stage.[14] In fact, this characteristic resulted in the Notani system classifying 13.2% of patients as having the most severe form of disease. In addition, the Notani system neglects to include the maxilla, a region where ORN is known to occur.[34] The Shaw system also performed well, predictable given that it is essentially the Notani system where some stage 1 cases are recategorized as minor bony spicules. The concept of minor bony spicules has been included is several previous classification systems[9,10] and represents an entity distinct from ORN, known to occur in 0.2% of healthy patients after third molar extraction.[35]

The CTCAE system performed well, as it characterizes ORN according to treatment need. As a result, this classification system is influenced by local availability and paradigms of treatment. In addition, Grade 4 ORN—life-threatening consequences, and Grade 5 ORN—death due to ORN, were not observed to occur in this study. Despite this, the CTCAE system remains the preferred grading system for clinical trials, which present a unique challenge. Those recording outcomes in trials may not be dental specialists and therefore ill-equipped to probe for exposed bone, limiting the detecting of stage I disease. Clinical trials favor medical imaging as opposed to dental plain films. Where computed tomography scans are used to assess response to therapy and monitor for recurrence, scatter from metallic restorations can significantly impede interpretation of regions surrounding teeth, making it difficult for medical imaging to diagnose Stage 0 disease. For these reasons, we harmonized the ClinRad system with the favored clinical trials system, the CTCAE, to provide a grading system for use in clinical trials (Table 5).

The AAOMS system for MRONJ was heavily consulted when developing the ClinRad system, as the AAOMS system is used almost universally by dental professionals to classify MRONJ. As a result, the system is used to stage MRONJ in real time, and there is overt harmonization across studies on MRONJ. This has allowed for the creation of high-quality evidence to guide decision making in patients at risk of MRONJ, an outcome we hope the ClinRad system will achieve.[32,36] To encourage uptake of a single system, we felt it important to incorporate characteristics favored by both medical and dental professionals involved in management of ORN. By aligning the ClinRad system with features of the AAOMS system, this makes the ClinRad system familiar to dentists, in particular oral and maxillofacial surgeons. Although most medical professionals will not be equipped with a dental probe and are unable to detect some cases of Stage I disease, ORN can still be effectively staged according to common radiographic appearance, including subtle changes on dynamic contrast-enhanced magnetic resonance imaging.

Notably, the system does not include duration of exposure, size of bone exposure, symptoms, or response to treatment. Analysis of systems that included a time or size component generally resulted in overestimation of severity of ORN, higher missingness, and nonsignificant performance on univariable analysis. Although patient symptoms will dictate treatment, this is poorly correlated with the extent of bone necrosis, and we instead tied symptoms to management suggestions. Finally, response to therapy requires a patient to have completed a therapy before a stage can be generated, limiting its utility in treatment decision making and standardized data capture.

Although the missingness for the ClinRad system was zero, this is not surprising as (1) the system was designed using existing systems with low missingness and (2) it was tested

and adjusted using patient cases with high missingness across various systems. The ClinRad system categorized 5.7% of patients with ORN as having the most severe form. This is similar to the proportion in systems that classify ORN in terms of functional deficits or required therapy, such as CTCAE (6.6%). Systems that categorize spread of ORN beyond the inferior alveolar canal (Notani et al[14]—13.2%, Shaw et al[16]—11.6%) appeared to overestimate the proportion of patients with severe ORN, while simple binary systems or those with high missingness had artificially elevated results.

When analyzing separability, Kaplan-Meier analysis demonstrated that the ClinRad system performed as well as the Notani et al,[14] Shaw et al,[16] RTOG/EORTC,[24] and Støre and Boysen[26] and Store systems. Only Schwartz and Notani systems required no grouping of stages for statistical power. Given the high proportion of missingness for Karagozoglu and Schwartz systems, the results for these systems should be interpreted with caution.

Our study has several limitations. The identified risk factors for development of ORN are based on a single-center retrospective review. The prescribed IMRT dose was incorporated in the ORN risk score model; however, the details on the delivered rather than planned IMRT dose were not available to be added. As expected, the ClinRad system performed well when applied to data used to develop the system but has not been tested outside of this retrospective data set. In addition, although Cox models demonstrated good concordance, time-dependent AUCs had wide confidence intervals and c-indices were not high, potentially because of the relatively low number of post-ORN jaw fracture or surgical intervention events. Future studies will be needed to demonstrate the performance in a prospective clinical setting and data sets from outside of our institution.

In conclusion, our study identified a high-risk group for ORN and proposed the ClinRad system that classifies ORN on the basis of clinical and radiographic findings. The ClinRad system outperformed existing systems in depicting the seriousness of ORN and may facilitate clinical care and clinical trials.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Disclosures provided by the authors are available with this article at DOI https://doi.org/10.1200/JCO.23.01951.

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ACKNOWLEDGMENT

The authors acknowledge the generous support by the Bartley-Smith/Wharton, the Gordon Tozer, the Wharton Head and Neck Translational, Dr Mariano Elia, the Joe's Team, and the Petersen Funds at the Princess Margaret Foundation of author's (S.H.H.) academic activity. The authors acknowledge the efforts of the dental summer research students, Linnaea Halpert, Josh Moyal, Mariam Ghobrial, Ahuva Plonka, Lila Shapiro, and Afisu Oladega.

REFERENCES

1. Singh A, Huryn JM, Kronstadt KL, et al: Osteoradionecrosis of the jaw: A mini review. Front Oral Health 3:980786, 2022
2. Studer G, Bredell M, Studer S, et al: Risk profile for osteoradionecrosis of the mandible in the IMRT era. Strahlenther Onkol 192:32-39, 2016
3. Zhang W, Zhang X, Yang P, et al: Intensity-modulated proton therapy and osteoradionecrosis in oropharyngeal cancer. Radiother Oncol 123:401-405, 2017
4. Moon DH, Moon SH, Wang K, et al: Incidence of, and risk factors for, mandibular osteoradionecrosis in patients with oral cavity and oropharynx cancers. Oral Oncol 72:98-103, 2017
5. Ben-David MA, Diamante M, Radawski JD, et al: Lack of osteoradionecrosis of the mandible after intensity-modulated radiotherapy for head and neck cancer: Likely contributions of both dental care and improved dose distributions. Int J Radiat Oncol Biol Phys 68:396-402, 2007
6. De Felice F, Tombolini V, Musio D, et al: Radiation therapy and mandibular osteoradionecrosis: State of the art. Curr Oncol Rep 22:89, 2020
7. Sawhney R, Ducic Y: Management of pathologic fractures of the mandible secondary to osteoradionecrosis. Otolaryngol Head Neck Surg 148:54-58, 2013

8. Park L, Lilic N, Addison B, et al: Cost analysis of osteoradionecrosis. J Laryngol Otol 131:303-308, 2017
9. Coffin F: The incidence and management of osteoradionecrosis of the jaws following head and neck radiotherapy. Br J Radiol 56:851-857, 1983
10. Morton ME, Simpson W: The management of osteoradionecrosis of the jaws. Br J Oral Maxillofac Surg 24:332-341, 1986
11. Clayman L: Clinical controversies in oral and maxillofacial surgery: Part two. Management of dental extractions in irradiated jaws: A protocol without hyperbaric oxygen therapy. J Oral Maxillofac Surg 55:275-281, 1997
12. Caparrotti F, Huang SH, Lu L, et al: Osteoradionecrosis of the mandible in patients with oropharyngeal carcinoma treated with intensity-modulated radiotherapy. Cancer 123:3691-3700, 2017
13. Schwartz HC, Kagan AR: Osteoradionecrosis of the mandible: Scientific basis for clinical staging. Am J Clin Oncol 25:168-171, 2002
14. Notani K, Yamazaki Y, Kitada H, et al: Management of mandibular osteoradionecrosis corresponding to the severity of osteoradionecrosis and the method of radiotherapy. Head Neck 25:181-186, 2003
15. Karagozoglu KH, Dekker HA, Rietveld D, et al: Proposal for a new staging system for osteoradionecrosis of the mandible. Med Oral Patol Oral Cir Bucal 19:e433-e437, 2014
16. Shaw R, Tesfaye B, Bickerstaff M, et al: Refining the definition of mandibular osteoradionecrosis in clinical trials: The cancer research UK HOPON trial (Hyperbaric Oxygen for the Prevention of Osteoradionecrosis). Oral Oncol 64:73-77, 2017
17. Glanzmann C, Grätz KW: Radionecrosis of the mandibula: A retrospective analysis of the incidence and risk factors. Radiother Oncol 36:94-100, 1995
18. Tsai CJ, Hofstede TM, Sturgis EM, et al: Osteoradionecrosis and radiation dose to the mandible in patients with oropharyngeal cancer. Int J Radiat Oncol Biol Phys 85:415-420, 2013
19. Lyons A, Osher J, Warner E, et al: Osteoradionecrosis—A review of current concepts in defining the extent of the disease and a new classification proposal. Br J Oral Maxillofac Surg 52:392-395, 2014
20. Marx RE: A new concept in the treatment of osteoradionecrosis. J Oral Maxillofac Surg 41:351-357, 1983
21. Wong K, Huang SH, O'Sullivan B, et al: Point-of-care outcome assessment in the cancer clinic: Audit of data quality. Radiother Oncol 95:339-343, 2010
22. Watson E, Eason B, Kreher M, et al: The DMFS160: A new index for measuring post-radiation caries. Oral Oncol 108:104823, 2020
23. Caton JG, Armitage G, Berglundh T, et al: A new classification scheme for periodontal and peri-implant diseases and conditions—Introduction and key changes from the 1999 classification. J Periodontol 89:S1-S8, 2018
24. Cox JD, Stetz J, Pajak TF: Toxicity criteria of the Radiation Therapy Oncology Group (RTOG) and the European Organization for Research and Treatment of Cancer (EORTC). Int J Radiat Oncol Biol Phys 31:1341-1346, 1995
25. LENT SOMA tables. Radiother Oncol 35:17-60, 1995
26. Støre G, Boysen M: Mandibular osteoradionecrosis: Clinical behaviour and diagnostic aspects. Clin Otolaryngol Allied Sci 25:378-384, 2000
27. National Cancer Institute: Common Terminology Criteria for Adverse Events (CTCAE), Version 5.0, 2017. https://ctep.cancer.gov/protocoldevelopment/electronic_applications/docs/ctcae_v5_quick_reference_5x7.pdf
28. Ruopp MD, Perkins NJ, Whitcomb BW, et al: Youden Index and optimal cut-point estimated from observations affected by a lower limit of detection. Biom J 50:419-430, 2008
29. Bansal A, Heagerty PJ: A tutorial on evaluating the time-varying discrimination accuracy of survival models used in dynamic decision making. Med Decis Making 38:904-916, 2018
30. Kamarudin AN, Cox T, Kolamunnage-Dona R: Time-dependent ROC curve analysis in medical research: Current methods and applications. BMC Med Res Methodol 17:53, 2017
31. Ruggiero SL, Dodson TB, Fantasia J, et al: American Association of Oral and Maxillofacial Surgeons position paper on medication-related osteonecrosis of the jaw—2014 update. J Oral Maxillofac Surg 72:1938-1956, 2014
32. Ruggiero SL, Dodson TB, Aghaloo T, et al: American association of oral and maxillofacial surgeons' position paper on medication-related osteonecrosis of the jaws—2022 update. J Oral Maxillofac Surg 80:920-943, 2022
33. Verduijn GM, Sijtsema ND, van Norden Y, et al: Accounting for fractionation and heterogeneous dose distributions in the modelling of osteoradionecrosis in oropharyngeal carcinoma treatment. Radiother Oncol 188:109889, 2023
34. Shokri T, Wang W, Vincent A, et al: Osteoradionecrosis of the maxilla: Conservative management and reconstructive considerations. Semin Plast Surg 34:106-113, 2020
35. Bui CH, Seldin EB, Dodson TB: Types, frequencies, and risk factors for complications after third molar extraction. J Oral Maxillofac Surg 61:1379-1389, 2003
36. Yarom N, Shapiro CL, Peterson DE, et al: Medication-related osteonecrosis of the jaw: MASCC/ISOO/ASCO clinical practice guideline. J Clin Oncol 37:2270-2290, 2019

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Development and Standardization of an Osteoradionecrosis Classification System in Head and Neck Cancer: Implementation of a Risk-Based Model

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Speakers' Bureau: Elekta
Research Funding: Elekta (Inst), RaySearch Laboratories (Inst), Oncospace (Inst)
Patents, Royalties, Other Intellectual Property: USPTO 11730561-Apparatus and methods for three-dimensional printed oral stents for head and neck radiotherapy [licensed to Kallisio, Inc] (Inst)
Travel, Accommodations, Expenses: Elekta, Philips Healthcare, Varian Medical Systems (Inst), Siemens Medical Solutions USA, Inc
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No other potential conflicts of interest were reported.

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APPENDIX

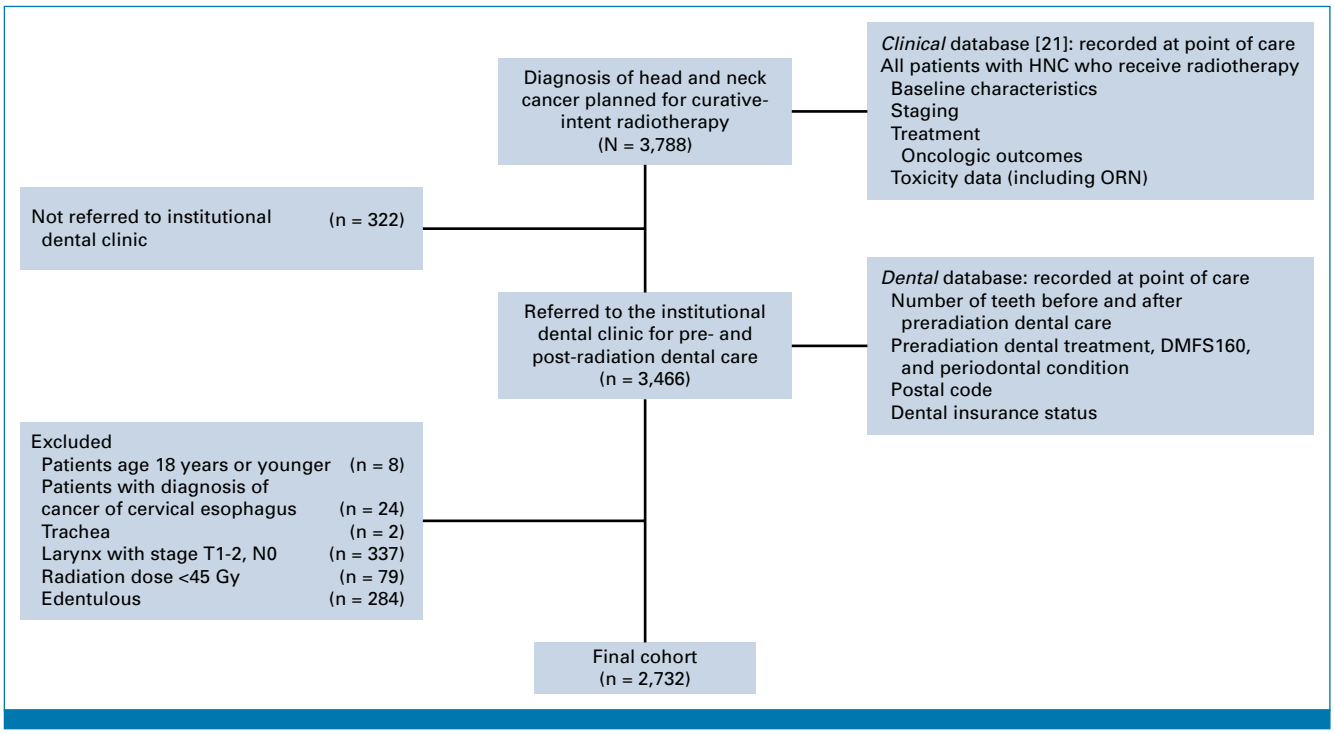

FIG A1. Flow diagram. DMFS, decayed, missing, filled surfaces according to DMFS160; HNC, head and neck cancer; ORN, osteoradionecrosis.

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TABLE A1. Summary of Classification Systems for Osteoradionecrosis

Classification	Author	Author Info	Year of Publication	Stages	Stages	Description	Basis of Stage	Maxilla and Mandible	Strength	Limitation
Coffin[9]	Frank Coffin	Oral surgeon	1983	2	Minor	A series of small sequestra that separate spontaneously after varying periods of weeks or months. These areas cannot be demonstrated radiologically	Clinical and radiologic findings	Both	Easy to implement	Subjective definition including varying periods of weeks or months. Two extremes, limited description of progression of ORN
					Major	Necrosis occurring to an extent that involves the entire thickness of the jaw, and a pathologic fracture is inevitable. This form can be obviously seen radiologically				
Marx[20]	Robert Marx	Oral surgeon	1983	3	I	Exposed alveolar bone without pathologic fracture, which responds to hyperbaric oxygen therapy	Clinical findings and treatment (HBOT, sequestrectomy, saucerization, resection, and reconstruction)	Both	Response from HBOT	Limited to cases where HBOT is used except stage III
					II	Disease does not respond to HBOT, and requires sequestrectomy and saucerization				
					III	Full-thickness bone damage or pathologic fracture, usually requires complete resection and reconstruction with free tissue				
Morton and Simpson[10]	M E Morton	Oral surgeon	1986	3	Minor	Ulceration with exposed bone and a history of bony spicules that healed spontaneously over a period of month	Clinical findings and response to treatment (sequestrectomy). Timing of exposed bone	Both	Able to tell if patient healed from sequestrectomy or not	Subjective definitions including a period of month, large areas of exposed bone, large sequestra. Emphasis on healed and healing, which are not always the case with ORN sequestra
	W Simpson	Oral surgeon			Moderate	Exposed bone and small sequestra limited in nature and healing spontaneously with conservative treatment within 6-12 months				
					Major	Large areas of exposed bone, with formation of large sequestra, possible fracture, and sinus formation. These cases often require radical treatment				
RTOG/EORTC[24]		Radiation oncologist	1995	5	0	No changes from baseline	Radiographic findings. Patient symptoms	Both	Wide application, however, because of nondescriptive definitions	Subjective definitions, no clear objective description of progression of ORN
					I	Asymptomatic; no growth retardation; reduced bone density				
					II	Moderate pain or tenderness; growth retardation; irregular bone sclerosis				
					III	Severe pain or tenderness; complete arrest of bone growth; dense bone sclerosis				
					IV	Necrosis/Spontaneous fracture				

(continued on following page)

TABLE A1. Summary of Classification Systems for Osteoradionecrosis (continued)

Classification	Author	Author Info	Year of Publication	Stages	Stages	Description	Basis of Stage	Maxilla and Mandible	Strength	Limitation
Glanzmann and Grätz[17]	Ch. Glanzmann	Radiation oncologist	1995	5	I	Bone exposure without signs of infection and persisting for at least 3 months	Clinical findings and response to treatment (mandibular resection). Timing of exposed bone	Mandible	Only stage that includes response from mandibulectomy	Limited descriptions on ORN progression in stage I and II. Death from ORN unnecessary. No mention of maxillectomy
	K.W. Grätz	Oral surgeon			II	Bone exposure with signs of infection or sequester and without the signs of grade 3-5				
					III	Bone necrosis treated with mandibular resection with a satisfactory result				
					IV	Bone necrosis with persisting problems despite mandibular resection				
					V	Death from ORN				
Clayman[11]	Lewis Clayman	Oral surgeon	1997	2	I	Bone lysis occurs under intact gingiva or mucosa	Clinical findings	Both	Includes radiographic ORN	Lack of description of ORN progression, no mention of the severe form of ORN (ie, path fracture, type 2 is too broad)
					II	A more aggressive type in which soft tissues break down, exposing the bone to saliva, and causing secondary contamination				
LENT/SOMA[25]		Radiation oncologist	1997		I	Unable to assess, mandibular radiograph (questionable changes or none)	Clinical and radiographic findings. Treatments required. Size of exposed bone	Mandible	More detailed than RTOG	Mandible only. Scoring method cannot be applied in our analysis. Different indication of treatments (ie, sequestrectomy, debridement, HBOT)
					II	Exposed bone ≤2 cm; antibiotics; mandibular radiograph (radiolucent, radiodense)				
					III	Exposed bone >2 cm or limited sequestration; debridement; HBO				
					IV	Fracture, surgical intervention or resection				
						Note: In our analysis, subjective was not included and we did not score				
Støre and Boysen[26]	G. Støre	Oral surgeon	2000	4	0	Mucosa defect only	Clinical and radiographical findings	Both	Includes radiographic ORN	Lack of description of ORN progression, no mention of the severe form of ORN (ie, path fracture, too big gap between stage II and III)
	M. Boysen	Otolarynologist			I	Radiologic evidence of necrotic bone with intact mucosa				
					II	Positive radiologic findings with denuded bone intraorally				
					III	Clinically exposed radionecrotic bone, verified by imaging techniques, along with skin fistulae and infection				

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TABLE A1. Summary of Classification Systems for Osteoradionecrosis (continued)

Classification	Author	Author Info	Year of Publication	Stages	Stages	Description	Basis of Stage	Maxilla and Mandible	Strength	Limitation
Schwartz and Kagan[13]	Harry C. Schwartz	Otolarynologist/oral surgeon	2002	5	I	Superficial involvement of the mandible only. Soft tissue ulceration is minimal. Only the exposed cortical bone is necrotic	Clinical and radiographical findings	Mandible	Detailed progression of ORN	Does not include radiographic ORN with intact mucosa
	A Robert Kagan	Radiation oncologist			II	Localized involvement of the mandibular cortex and underlying medullary bone				
					IIa	Soft tissue ulceration is minimal				
					IIb	Soft tissue necrosis including orocutaneous fistulation				
					III	Full thickness of a segment of bone is involved, including lower border. Pathologic fracture may occur				
					IIIa	Soft tissue ulceration is minimal				
					IIIb	Soft tissue necrosis including orocutaneous fistulation				
Notani et al[14]	Ken-ichi Notani	Oral surgeon	2003	3	I	ORN confined to mandibular bone	Clinical and radiographical findings	Mandible	Simple, easy, and objective	Mandible only. No descriptive definition of ORN
	Motoyasu Nakamura	Oral and maxillofacial radiology			II	ORN limited to the alveolar bone and/or mandible above the level of inferior alveolar canal				
					III	ORN involving the mandible below the level of inferior alveolar canal and/or skin fistula and/or pathologic fracture				
Tsai et al[18]	Chiaojung Jillian Tsai	Radiation oncologist	2012	4	I	Minimal bone exposure with conservative management only	Treatment required	Mandible	Show progression of more invasive/involving treatments required	Subjective definition including minimal bone exposure and conservative management. Not applicable to cases that did not receive treatments listed. Subjective criteria of when these treatments are implemented
	Theresa M. Hofstede	Dental oncologist			II	Minor debridement (sequestrectomy) received				
	Erich M. Sturgis	Otolarynologist			III	Hyperbaric oxygen needed				
					IV	Major surgery required (mandibulectomy or hemimandibulectomy, removal of necrotic bone from the mandible, laryngopharyngectomy, multiple dental extractions and 4-quadrant alveoloplasty, and curettage of abscesses)				

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TABLE A1. Summary of Classification Systems for Osteoradionecrosis (continued)

Classification	Author	Author Info	Year of Publication	Stages	Stages	Description	Basis of Stage	Maxilla and Mandible	Strength	Limitation
Karagozoglu et al[15]	Kemal H. Karagozoglu	Oral surgeon	2014	6	0	Exposure of mandibular bone for less than 1 month; no distinct changes on plain radiographs (panoramic radiograph or periapical film)	Clinical and radiographical findings. Duration of exposed bone	Mandible	Includes objective clinical and radiographic findings and subjective patient symptoms	Does not take into account exposed bone less than 1 month with radiographic changes, radiographic ORN
					I	Exposure of mandibular bone for at least 1 month; no distinct changes on plain radiographs (panoramic radiograph or periapical film). Asymptomatic otherwise, eg, no pain or presence of cutaneous fistulas (I A), or symptomatic, eg, pain or presence of cutaneous fistulas (I B)				
					II	Exposure of mandibular bone for at least 1 month; distinct changes present on plain radiographs (panoramic radiograph or periapical film), but not involving the lower border of the mandible. Asymptomatic otherwise, eg, no pain or presence of cutaneous fistulas (II A), or symptomatic, eg, pain or presence of cutaneous fistulas (II B)				
					III	Exposure of mandibular bone for at least 1 month; distinct changes on plain radiographs (panoramic radiograph or periapical film), involving the lower border of the mandible, irrespective of any other signs and symptoms				
						Note: In case of doubt about the presence and/or extent of radiographical bone involvement, the lower stage should be allotted				

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TABLE A1. Summary of Classification Systems for Osteoradionecrosis (continued)

Classification	Author	Author Info	Year of Publication	Stages	Stages	Description	Basis of Stage	Maxilla and Mandible	Strength	Limitation
Lyons et al[19]	Andrew Lyons	Oral surgeon	2014	4	I	< 2.5 cm length of bone affected (damaged or exposed); asymptomatic; medical treatment only	Clinical and radiographic findings. Size of exposed bone. Treatments required	Mandible	Includes objective clinical and radiographic findings and subjective patient symptoms	Does not take into account exposed bone < 2.5 cm that is symptomatic, radiographic ORN.
					II	>2.5 cm length of bone; asymptomatic; including pathologic fracture or involvement of inferior dental nerve or both; medical treatment only unless there is dental sepsis or obviously loose, necrotic bone				
					III	>2.5 cm length of bone; symptomatic, but with no other features despite medical treatment consider debridement of loose or necrotic bone, and local pedicled flap				
					IV	2.5 cm length of bone; pathologic fracture, involvement of inferior dental nerve, or orocutaneous fistula, or a combination reconstruction with free flap if patient's overall condition allows				
CTCAE v5.0[27]			2017	4	I	Asymptomatic; clinical or diagnostic observations only; intervention not indicated	Clinical findings and treatments indicated	Mandible	One of few systems that take symptoms into account	Lack of radiographic progression (ie, panorex). No clear definition of ORN (ie, exposed bone?), based on subjective findings (symptoms) and subjective indication of treatment (when to do elective operation?)
					II	Symptomatic; medical intervention indicated (eg, topical agents); limiting instrumental ADL				
					III	Severe symptoms; limiting self-care ADL; elective operative intervention indicated				
					IV	Life-threatening consequences; urgent intervention indicated				
					V	Death				

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TABLE A1. Summary of Classification Systems for Osteoradionecrosis (continued)

Classification	Author	Author Info	Year of Publication	Stages	Stages	Description	Basis of Stage	Maxilla and Mandible	Strength	Limitation
Caparrotti et al[12]	Francesca Caparrotti	Radiation oncologist	2017	5	I	Loss of mucosa with exposed bone requiring antibiotic or need for sequestrectomy for a duration of ≤ 3 months	Clinical findings and treatments indicated. Timing of exposed bone	Both	Show persistence of exposed bone needing antibiotics or sequestrectomy through time	Does not take into account radiographic ORN. Does not take into account exposed bone not requiring antibiotics or sequestrectomy. No clear indication of treatments (HBOT, pentoxyfylline/vitamin E, surgery). Death stage unnecessary
	Andrew Hope				II	Loss of mucosa with exposed bone requiring antibiotic or need for sequestrectomy for a duration of >3 months and ≤6 months				
					III	Loss of mucosa with exposed bone requiring antibiotic or need for sequestrectomy for a duration of >6 months, or need for hyperbaric oxygen therapy, or medical management with pentoxifylline/vitamin E				
					IV	Loss of mucosa with exposed bone requiring antibiotic or need for sequestrectomy for a duration of >12 months, or need for surgery, or pathologic bone fracture				
					V	Death				
Shaw et al[16]	Richard J. Shaw	ENT	2019		Mucosal healing	Healed mucosa and absence of radiographic changes	Clinical and radiographic findings	Mandible	Includes small sequestra that are not true ORN. Provides descriptive definition of ORN that was missing in Notani	Uses Notani system that is designed for mandible only
	Other authors	Oral surgeon			Minor bone spicules	Exposed bone measures <20 mm^2 and in the absence of radiographic changes				
					Notani 1	Exposed bone ≥20 mm^2 and radiographic changes, both confined to dentoalveolar process				
					Notani 2	Exposed bone ≥20 mm^2 and radiographic changes beyond dentoalveolar process but above inferior dental nerve canal				
					Notani 3	Exposed bone ≥20 mm^2 and radiographic changes affecting the full thickness of mandible, or pathologic fracture, or extraoral fistula				

Abbreviations: CTCAE, Common Terminology Criteria for Adverse Events; HBOT, hyperbaric oxygen therapy; ORN, osteoradionecrosis; RTOG, Radiation Therapy Oncology Group.